

Temporal-Spatial Taxonomy for Study Design

Temporal-Spatial Taxonomy for Study Design I

DigiPPPiP is organized by two primary axes: temporal structure and spatial configuration. The configured project has 3 temporal modes, 3 spatial configurations, and 9 total modality cells. [fig:taxonomy_matrix] is the central map of that design space. The axes intentionally separate relation from infrastructure: two partners can be relationally engaged while co-present, remote, delayed, or hybrid, but each cell changes what can be perceived, logged, repaired, and revisited.

Temporal-Spatial Taxonomy for Study Design II

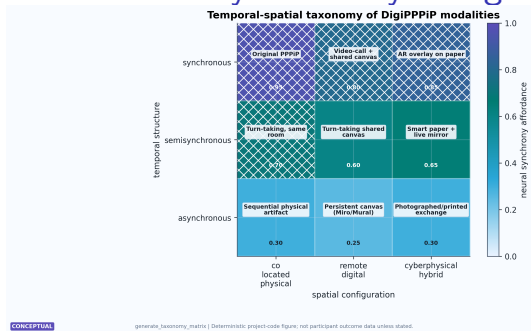


Figure 1: Three-by-three temporal-spatial taxonomy for DigiPPPiP. The matrix is generated by `generate_taxonomy_matrix()` from `src/taxonomy.py`; read rows as temporal structures, columns as spatial configurations, cell labels as modality names, numeric labels as conceptual neural-synchrony affordance scores, and hatching as high-affordance cells. It supports decision-making about study conditions rather than product ranking. The scores are design variables, and measured synchrony would require participant physiology and controls.

Temporal-Spatial Taxonomy for Study Design III

The taxonomy is not a typology for naming products. It is a decision procedure for matching relational need to temporal-spatial form. A study should first identify the hard constraint: co-presence, distance, disability access, uneven schedules, privacy risk, or place memory. It should then choose the simplest cell that satisfies that constraint while preserving agency and mutual interpretability. This prevents the framework from treating the most instrumented cell as the most advanced one.

Table 1: Temporal-spatial DigiPPPiP modality grid. The renderer numbers this table from the label rather than from prose.

Temporal structure	Co-located physical	Remote digital	Cyberphysical hybrid
Synchronous	Original PPPiP	Video-call plus shared canvas	AR overlay on paper
Semisynchronous	Turn-taking in the same room	Turn-taking shared canvas	Smart paper plus live mirror
Asynchronous	Sequential physical artifact	Persistent shared canvas	Photographed or printed exchange

Temporal-Spatial Taxonomy for Study Design IV

[@tbl:taxonomy_modes] should be read as a decision aid. A couple seeking immediate mutual attunement may choose synchronous co-presence. A long-distance couple with uneven schedules may choose asynchronous remote drawing. A pair interested in place memory may choose a cyberphysical hybrid that links physical marks with a persistent digital archive. The highest conceptual neural-synchrony score in the configured taxonomy is 0.95, but no single cell is universally best. The appropriate cell depends on access needs, privacy constraints, device availability, emotional goals, and the amount of persistence a dyad actually wants.